EE 446 TinyML
Homework 2 Writeup

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Question 2: Dimensionality Reduction for Visualization

Attack points are red and normal points are blue in all three figures. t-SNE and Kernel PCA were computed on a random subsample of 2,000 test points for tractability.

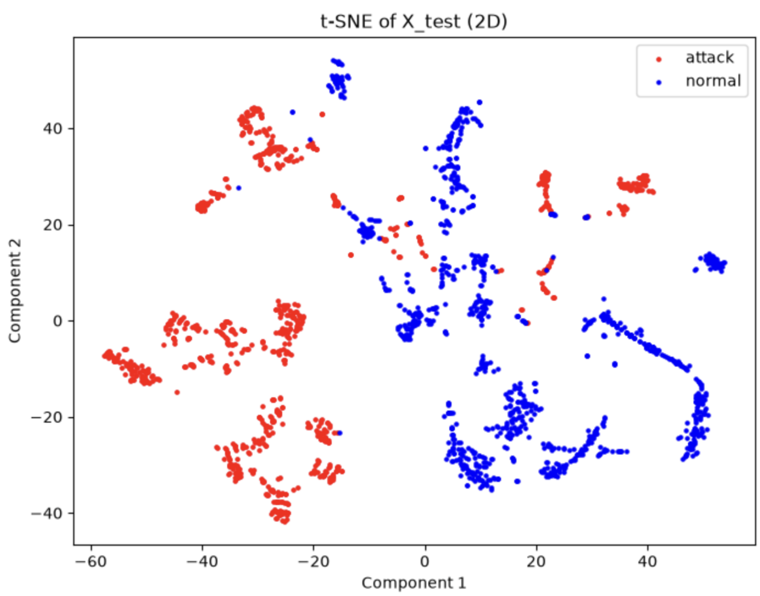

Figure 1: Question 2(a): t-SNE projection of the test set in 2D.

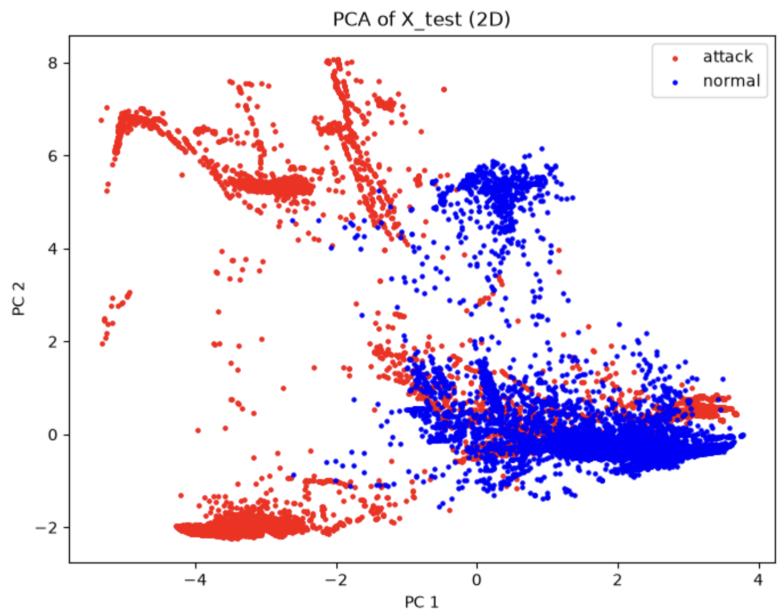

Figure 2: Question 2(b): PCA projection of the test set in 2D.

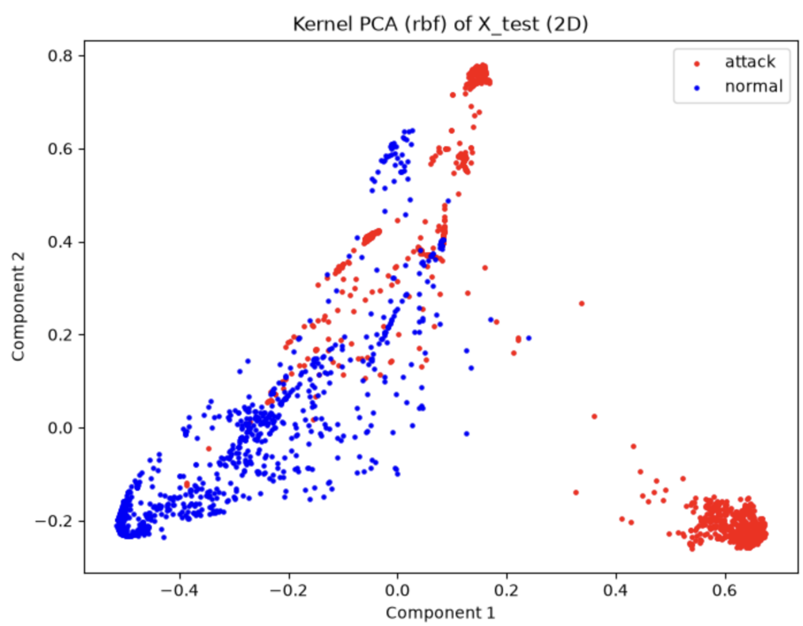

Figure 3: Question 2(c): Kernel PCA (rbf) projection of the test set in 2D.

Question 3: Baseline DNN

```
788/788 [==============================] - 0s 230us/step
              precision    recall  f1-score   support

      attack     0.9980    0.9984    0.9982     11736
      normal     0.9986    0.9982    0.9984     13459

    accuracy                         0.9983     25195
   macro avg     0.9983    0.9983    0.9983     25195
weighted avg     0.9983    0.9983    0.9983     25195

[[11717    19]
 [   24 13435]]
```

Figure 4: Question 3(c): classification report and confusion matrix for `base_model` on the test set.

The float32 model reaches 0.9983 accuracy on 25,195 test samples, with 19 attacks predicted as normal and 24 normal connections predicted as attacks.

Question 4: Full-Integer Quantized Model

The converter reported an input tensor scale of 0.09267 with zero point −36, and an output tensor scale of 0.00390625 with zero point −128.

```
Original model size: 51.80 KB
Quantized model size: 4.30 KB
```

Figure 5: Question 4(a): original and quantized model file sizes.

```
input scale/zp: 0.09267217665910721 -36
output scale/zp: 0.00390625 -128
              precision    recall  f1-score   support

      attack     0.9977    0.9986    0.9982     11736
      normal     0.9988    0.9980    0.9984     13459

    accuracy                         0.9983     25195
   macro avg     0.9983    0.9983    0.9983     25195
weighted avg     0.9983    0.9983    0.9983     25195

[[11720    16]
 [   27 13432]]
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
```

Figure 6: Question 4(b): classification report and confusion matrix for `tflite_quant_model` on the test set.

The quantized model also reaches 0.9983 accuracy, with 16 attacks predicted as normal and 27 normal connections predicted as attacks, while the model size drops from 51.80 KB to 4.30 KB.

Question 5: Deployment on the Arduino Nano 33 BLE Sense

```
16:47:42.058 -> Sample #0 | Predicted Class: 1 | Actual Class: 1
16:47:43.079 -> Sample #1 | Predicted Class: 0 | Actual Class: 0
16:47:44.066 -> Sample #2 | Predicted Class: 0 | Actual Class: 0
16:47:45.089 -> Sample #3 | Predicted Class: 1 | Actual Class: 1
16:47:46.080 -> Sample #4 | Predicted Class: 1 | Actual Class: 1
```

Figure 7: Question 5(c): serial monitor output for the first five test samples.

```
17:02:46.403 -> Sample #0 | Predicted Class: 0 | Actual Class: 0
17:02:47.394 -> Sample #1 | Predicted Class: 1 | Actual Class: 1
17:02:48.382 -> Sample #2 | Predicted Class: 1 | Actual Class: 1
17:02:49.404 -> Sample #3 | Predicted Class: 0 | Actual Class: 0
17:02:50.392 -> Sample #4 | Predicted Class: 1 | Actual Class: 1
17:02:51.411 -> Sample #5 | Predicted Class: 1 | Actual Class: 1
17:02:52.402 -> Sample #6 | Predicted Class: 1 | Actual Class: 1
17:02:53.392 -> Sample #7 | Predicted Class: 0 | Actual Class: 0
17:02:54.414 -> Sample #8 | Predicted Class: 0 | Actual Class: 0
17:02:55.403 -> Sample #9 | Predicted Class: 0 | Actual Class: 0
```

Figure 8: Question 5(d): serial monitor output for the ten test samples following the first five.

All fifteen on-device predictions match their actual labels.